

Wei Liu
Full Stack Developer/React JS Developer
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SUMMARY

I have 8 years' experience in software industry in designing, developing, creating, and testing web applications using **JavaScript, NodeJS, React/Redux, HTML5/CSS3, GraphQL** and **RESTful APIs**.

- Designed several single-page and multi-page web applications using JavaScript with **NodeJS** and **React** framework.
- Proficient in **JavaScript**, including the **DOM manipulation** of web pages. Experience with common JavaScript development tools such as **Babel, Webpack** and **npm**.
- Proficient in using different types of databases, such as **MySQL** and **NoSQL** databases, such as **MongoDB**, in the applications to manage needed data and files. Knowledge in Object-relational mapping framework, such as **Hibernate**, to integrate relational databases into the applications.
- Created the front-end part of the web applications using **React** framework with **Redux** as the state management tool of React components. Used **Redux Saga** in the projects to handle asynchronous request and response between the server and the front-end of web applications.
- Created responsive user interfaces of web applications efficiently with **Material-UI** and **Bootstrap** library. Implemented real-time communication between the front-end and back-end of web applications with the **Socket.IO** library.
- Developed applications with React components quickly and more flexible using **React Hooks**. Used **React Router** to switch the web view from different React components.
- Experience in developing multi-page web applications with **SpringMVC** framework and **React** components.
- Experience in **RESTful** web services to integrate the information between server side to client side of applications.
- Tested the web applications using **Jasmine** framework. For developing large scale applications, used **Scrum** and **Jira** to manage the process of project development.
- Hands-on experience of major cloud computing services, such as **Amazon AWS** and **Google Cloud Platform**, to switch the database logic and application deployment from the traditional server to cloud. Experience in pack, ship and run applications in the containers with **Docker** and **Kubernetes**.
- Proficient in software development at different operating systems, such as **MacOS, Windows** and **Linux**. Proficient in using **Git** or **SVN** for version control when develop large-scale applications.
- Experience in object-oriented programming with different programming languages, such as **Python, Java**, and **C++**.
- Have good communication skills. Could cooperate with engineers in a large team to develop a large-scale application.

TECHNICAL SKILLS

- **Languages:** JavaScript/TypeScript, Java, C/C++, Python, SQL, HTML5/CSS3
- **Frameworks:** React/Redux, GraphQL/Apollo, ExpressJS, Spring Boot, Socket.IO, Bootstrap, Material-UI, WebGL, Hadoop, Spark, Ant Design
- **Tool & Software:** AWS, Google Cloud Platform, Docker, Kubernetes, Git, Jasmine, Jest, Postman
- **Databases:** MySQL, MongoDB, Redis, AWS S3
- **OS:** MacOS, Windows, Linux
- **Project Management Tools:** Jira, Scrum

EDUCATION

The Ohio State University

Master of Science, Computer Science and Engineering

Columbus, Ohio

Dec. 2020

WORKING EXPERIENCE

eBay Inc.

Position: Software Engineer

Bellevue, Washington

July 2023 – Present

Project: Email Campaign Manager

Description:

Working with a team of engineers to develop and maintain the web application and internal services for the Email Campaign Management system. The platform provides a user interface for marketing teams to create, schedule, and track campaigns, visualizing real-time engagement metrics through integrated service layers. The system manages the lifecycle of marketing campaigns across marketing email and app push notifications, handling a volume of over 440 million emails sent per week. The platform integrates with internal targeting engines and external delivery providers to support complex experimentation and real-time reporting.

Responsibilities:

- Worked on an agile software development team to build modern, scalable and robust system and web application.
- Migrated the legacy system from unmanaged MySQL and bare-metal servers to new architecture using **microservices** and managed **Oracle databases** to improve scalability and fault tolerance.
- Developed the Campaign Manager web application using **React**, providing an intuitive interface for marketing teams to create, update, and search campaigns across Email and Push Notification channels.
- Managed the web application backend services using **NodeJS**, responsible for serving the front-end artifacts and orchestrating API calls for the internal campaign management tools.
- Implemented a **GraphQL** service layer to aggregate data from multiple backend services and MySQL databases, creating a unified schema for the front-end to consume campaign metadata and templates.

- Implemented a real-time tracking pipeline to capture delivery metrics (via **SendGrid webhooks**) and engagement metrics (clicks/opens via **Jetstream** and **Rover**), aggregating data into MySQL for reporting.
- Optimized the data fetching strategy for the delivery metrics, using **GraphQL** queries to retrieve granular delivery and engagement metrics (bounced, delivered, unsubscribed) from the underlying user-level data files and databases.
- Engineered real-time data visualization features in the **React dashboard** to display near real-time email engagement metrics (opens/clicks), fetching aggregated stats from the backend.
- Implemented a robust event-based alerting system to notify campaign managers of lifecycle states (Pre-Start, During-Run, Post-Start) and errors using rendered email templates.
- Integrated **Kafka** platform for asynchronous job queuing and inter-service communication, ensuring durable job processing and automatic partition re-assignment during worker node failures.

Palo Alto Networks
Position: Software Engineer

Santa Clara, California
 May 2022 – Mar. 2023

Project: Customer Support Portal

Description:

Working with a team of engineers to develop the customer support portal. With the support portal, customers could register and manage network security devices and services purchased from the company. Developed the front-end applications with React/Redux and back-end services with Java and NodeJS for the system.

Responsibilities:

- Created the front-end part of the system based on **JavaScript/TypeScript, HTML5/CSS3, React** and **Redux**.
- Refactored the outdated device management pages using the new stack **React, Redux and React Router**.
- Divided the front-end UI of the system to various **micro frontends** to simplify the development and test process. Developed small micro frontends with **React** and **npm** to build large and complex pages.
- Developed, tested and maintained shareable and re-usable UX components built with **React** and **TypeScript**. Designed and developed UI component libraries with **Storybook**.
- Developed **React** components with **styled-components** library to simplify the style handling in React applications and improve performance of complicated pages.
- Used **Redux** to manage the state and data needed for the front-end of the application. Used **React Hooks** to manage the local state needed in some React Components. Refactored old Redux code and back-end calls with **Redux Toolkit** to simplify the code and improve front-end efficiency.
- Integrated the authentication system to the web application with the lifecycle of React components. Used Redirect links to prevent unauthorized CRUD operations.
- Designed some RESTful APIs for retrieving the information of network security services on the back end of the system with **NodeJS** library to manage the information stored in the database.

- Maintained **MongoDB** database for fast read, write, and query operations in some backend services for the network security device management. Created the data schemas with **Mongoose** library in front-end applications to manipulate the device information stored in the **MongoDB** database.
- Worked with teammates to deploy the system to **Google Cloud Platform**. Tested the frontend of the system using **Jasmine** framework. Added automated browser tests for the front-end application using **BrowserStack**.

Expedia Group

Position: Full Stack Software Engineer

Seattle, Washington
Sept. 2021 – May 2022

Project: Property Onboarding Platform for Lodging

Description:

Developing the onboarding system with a team of engineers for property onboarding. Hotels and house owners could use this system to onboard their properties to the internal system of Expedia for customer booking. The system consists of node and react applications to guide partners onboarding new properties and GraphQL services to process the onboarding request.

Responsibilities:

- Migrated the old onboarding feature from **Redux / RESTful APIs** to **React Hooks / GraphQL** services.
- Implemented custom, and self-validating form components using **React Hooks, Formik, and Yup**.
- Developed flexible and reliable **custom hooks** to extract common logic from different components
- Used **Apollo Client** to generate data models and process **GraphQL** operations for query and mutation of the data getting from the back-end services.
- Developed a federated GraphQL service to process query and mutation operations. Implemented resolvers in the service to populate the data from the backend service and generate the response for the GraphQL requests.
- Utilized the **server driven UI** approach to create components for some pages in the onboarding application. Developed backend APIs to describe which components are part of a screen and their order. Then front-end apps render their native implementation of all the components according to GraphQL query response of back-end services.
- Used **webpack** to bundle **JavaScript** files of the web application for usage in browsers. Used **Git** for the version control of source code and **npm** for **JavaScript** package management to develop the front end of the system. Adopted **ESLint** to manage **JavaScript** codes to unify the style of coding, increase readability and maintainability of the code base and find bugs before executing the code.
- Implemented the unit-test of the front-end application using **Jest** framework. Developed and tested GraphQL services with the **Postman** and **Apollo Studio**.
- Collaborated with other team members to optimize and develop the system and worked on **Jira** for managing the development tasks and bug fixes.

Blue Cross Blue Shield
Position: React Developer

Columbus, Ohio
Sept. 2020 – Aug. 2021

Project: Doctor Management and Search System

Description:

Worked on developing the full stack web application for the doctor management and search with a team of engineers. With the management system, the company could manage the information of doctors in different regions in the country. Customers could find doctors nearby with their zip code or location information.

Responsibilities:

- Designed the homepage, doctor information page, search nearby doctor page and the doctor information edit page with **JSX**, **XML** and **HTML5/CSS3**.
- Worked with other engineers in the team to design and develop exciting features using **JavaScript** language.
- Utilized **Redux Thunk** middleware to handle asynchronous request and response between the server and the front-end of the application.
- Developed the customer service component of the system using **React Hooks** and **Socket.io** library to provide automatic response and real-time communication between customers and agents.
- Designed the **RESTful APIs** on the back end with **ExpressJS** framework to allow adding, updating, and deleting the information of doctors stored in the databases.
- Utilized **Google Map APIs** to get the location information of users and search doctors nearby with the zip code in the database.
- Utilized **NoSQL** database to store the information of doctors in the project. Used **MongoDB** documents to store the information of doctors. Designed the data schemas of the system with **Mongoose** library and mapped the data schemas to MongoDB documents. Used the schemas to manage the data collections of users to do adding, updating, deleting, and searching operations to the documents to manage the information of doctors and search nearby doctors.
- Collaborated with other engineers to deploy the web application on the **Amazon AWS EC2** instances.

Wells Fargo
Position: Full Stack Developer

Columbus, Ohio
Jan. 2019 - Aug. 2020

Project: Auto Loans Management System

Description:

Worked with a team of engineers to develop the system of loan management for cars. The system focused on the services of auto loans. Customers could use this system to set up payment methods and check the information and monthly statements of their loans.

Responsibilities:

- Designed and developed the front-end pages of the system, such as the information page and statement browse page, using **JavaScript**, **HTML5/CSS3** with **Ajax** controls, **JSON** and **XML**.
- Revamped and optimized the UI, UX of the system with libraries including **Bootstrap** and **Material-UI**.
- Developed the payment method setup page using **JavaScript** and event handling methods via the manipulation of the **DOM model**.
- Extensively implemented React components, props, forms, events, keys and **Flux** concept.
- Used **React Router** to provide user-friendly experience when switch from different pages. Used **Redux** to manage the state and information of needed by different React components.
- Worked with **Redux Thunk** middleware and **React** framework in the front-end of the system to retrieve and store data from **MongoDB** database with the **RESTful APIs** provided the back end of the application.
- Worked closely with the back-end team to display the loan information of customers using the custom **React** components and **Redux**.

BookKuu.com

Position: Software engineer

Hangzhou, China

Jan. 2014 - July 2017

Project: Online Bookstore (Web Application of Book Purchase)

Description:

Developed the full-stack web application for selling books online with a large team of engineers. The web application provided user-friendly experience for users to browse, select and buy books online from the company. It also enabled the sales teams to capture data and get instant insights to manage the stocks of books.

Responsibilities:

- Used **SpringMVC** framework and **Java Servlets** for the back end of the application to receive request and send responses from and to front-end of the web application. Used **MySQL** database to store the payment and order status information.
- Built the responsive front-end part of the web application with **React** and **Redux** for people to browse and select the books they want to buy.
- Implemented interactive web pages at the front-end part of the application with **Bootstrap**, **React Router**, **Ajax** and **MaterialUI** libraries and **HTML5/CSS3** to assemble responsive user interfaces.
- Used **Redux** as the central state management tool for the front-end of the application. Used **Redux Thunk** middleware to process asynchronous calls between the front-end and server side of the application.
- Developed a **RESTful** web service to process shipping information in the orders. Used **JMS** to notify user order status update.
- Utilized **JEST** as tool to test code for the front-end of the application to reduce the potential bugs in the application. Used **JTest** to test the back end of the application.
- Adopted **Docker** as containers for effective deployment of this web application.